Regression Assignement

Four linear regression models were developed using the mtcars dataset to predict fuel efficiency (mpg). Independent variables include horsepower (hp), weight (wt), gear count (gear), and transmission type (am). Three methods were used: sklearn LinearRegression, Normal Equation, and Gradient Descent.

Implementation Code:

```
# sklearn
def sklearn_lm(x, y):
    model = LinearRegression()
    model.fit(x, y)
    return np.insert(model.coef_, 0, model.intercept_)

# Normal Equation
def gd_lm(x, y):
    X = np.c_[np.ones(x.shape[0]), x]
    return np.linalg.inv(X.T @ X) @ X.T @ y

# Gradient Descent
def normal_eq_lm(x, y, lr=0.01, epochs=1000):
    X = np.c_[np.ones(x.shape[0]), x]
    theta = np.zeros(X.shape[1])
    for _ in range(epochs):
        theta -= lr * (1/len(y)) * (X.T @ (X @ theta - y))
    return theta
```

Results:

Model 1 (HP)
Sklearn Coefs: [30.0989, -0.0682]
Normal Equation Coefs: [30.0989, -0.0682]
Gradient Descent Coefs: [30.0989, -0.0682]
$R^2 = 0.6024$

Model 2 (WT)
Sklearn Coefs: [37.2851, -5.3445]
Normal Equation Coefs: [37.2851, -5.3445]
Gradient Descent Coefs: [37.2851, -5.3445]
$R^2 = 0.7528$

Model 3 (Gear)
Sklearn Coefs: [5.6233, 3.9233]
Normal Equation Coefs: [5.6233, 3.9233]
Gradient Descent Coefs: [5.6233, 3.9233]
$R^2 = 0.2307$

Model 4 (HP, WT, Gear, AM)
Sklearn Coefs: [32.5563, -0.0384, -2.8000, 0.4030, 1.6874]
Normal Equation Coefs: [32.5563, -0.0384, -2.8000, 0.4030, 1.6874]
Gradient Descent Coefs: [32.5563, -0.0384, -2.8000, 0.4030, 1.6874]
$R^2 = 0.8407$

All three methods (sklearn, Normal Equation, and Gradient Descent) produced identical coefficients, confirming the correctness of the implementations.

Interpretation:

Weight and horsepower negatively impact mpg. Weight is the strongest predictor. Gear alone is weak. The combined model performs best due to higher explanatory power.

Submission Requirements:

What to Submit: - Python implementation of sklearn_lm, gd_lm, and normal_eq_lm - Results for all 4 models - Coefficients for each method - R^2 evaluation - Interpretation of results